

EXPERIMENT 6

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EXPERIMENT: Advanced SQL Queries

AIM:

To learn and apply advanced SQL concepts such as conditional logic, aggregate functions, joins, self-joins, and date-related operations for effectively solving practical database problems.

Question 1:

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges. Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

Solution:

```
SELECT sale_date,  
       SUM(  
         CASE  
           WHEN fruit = 'apples' THEN sold_num  
           ELSE -sold_num  
         END  
       ) AS diff  
FROM Sales  
group by sale_date  
order by sale_date asc;
```

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Data Output	Messages	Notifications
Showing rows: 1 to 4		
	sale_date date	diff bigint
1	2020-05-01	2
2	2020-05-02	0
3	2020-05-03	20
4	2020-05-04	-1

Question 2:

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- id represents the current node.
- p_id represents the parent node.

Classify every node into one of the following categories:

- Root: A node that does not have any parent.
- Leaf: A node that has a parent but does not have any children.
- Inner: A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

Solution:

```
select id,
CASE
  when p_id is null then 'ROOT'
  when id in (SELECT p_id from Tree) then 'INNER'
  else 'LEAF'
end as type
from Tree
```

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Data Output Messages Notifications

Showing rows: 1 to 5

	id [PK] integer	type text
1	1	ROOT
2	2	INN...
3	3	LEAF
4	4	LEAF
5	5	LEAF

Accepted 20 / 20 testcases passed

Shagun Goyal submitted at Aug 17, 2026 02:10

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Runtime
507 ms | Beats 58.75%

```

1 # Write your MySQL query statement below
2 select id,
3 case
4 when p_id is null then 'Root'
5 when id in(select p_id from Tree) then 'Inner'
6 else 'Leaf'
7 end as type
8 from Tree
  
```

LEARNING OUTCOMES:

1. Learned how to apply self-joins to compare entries within the same table and find the difference between apples and oranges sold on a particular date.
2. Understood how JOIN conditions and WHERE clauses can be used to connect related records using common fields such as dates and fruit categories.
3. Gained knowledge of using the CASE statement to categorize hierarchical records as Root, Inner, or Leaf nodes according to their parent-child structure.
4. Learned how subqueries can be used to check parent-child relationships and identify whether a particular node has any children.
5. Improved my ability to combine JOIN, CASE, subqueries, conditional expressions, and ORDER BY to solve practical problems related to sales data and hierarchical structures.



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